Mohammad Ali Safaei

Email | Personal Website

Linkedin | Google Scholar

EDUCATION

University of Tehran – Tehran, Iran ***(#322 in QS Rankings)*** ***2021 – 2024***

M.Sc. in Mechanical Engineering |Applied Mechanics specialization

Thesis Title: Distinctive Electromechanical Performance of Metamaterial Dielectric Elastomer Actuators with TPMS Geometries [Link].

Supervisor(s): Dr. Mostafa Baghani, Dr. Majid Baniassadi.

GPA: 18.15/20.00.

Bu-Ali Sina University – Hamedan, Iran ***2016 - 2021***

B.Sc. in Mechanical Engineering

Thesis Title: Optimization of Energy Absorption in Foam-filled Sandwich Aluminum Tubes [Link].

Supervisor(s): Prof. Ali AlaviNia.

GPA: 15.66/20.00.

RESEARCH INTERESTS

- Computational methods (FEM)
- Multi-physics Constitutive Modeling
- Physics-informed Neural Networks
- Architected/lattice structures
- Inverse design, Generative AI
- 3D/4D printing

SKILLS

English proficiency: IELTS **Academic Overall** Band **7.5** ***(CEFR C1; L 7.0, R 9.0, W 7.0, S 6.5)***

Programming: Python (PyTorch, JAX, etc.), MATLAB, Fortran, JavaScript.

CAD software [Link]: SOLIDWORKS ,CATIA.

CAE software [Link]: ABAQUS, COMSOL Multiphysics, , LS-DYNA.

Other software: Simplify3D, Cura, LAMMPS, Materials Studio, MS Office (Word, PPT, Excel).

Hands-on Experiences: Casting, Welding, CNC Machining, Nondestructive testing (NDT).

PUBLICATIONS

M. A. Safaei, M. Baghani, M. Baniassadi, and M. Bodaghi, "In Silico Actuation Performance Investigation of Dielectric Elastomers with TPMS Geometries," *European Journal of Mechanics - A/Solids*, p. 105540, 2024 . [Link]

Safaei, M.A., Baniassadi, M., Baghani, M., Shahsavari, H. (2026). Toward Responsive Polymer Actuators Based on Novel Composite Structures. In: Bhaskar, S., Matsumi, N., Mozetič, M., John, M.J., Thomas, S. (eds) Smart Polymers for Sensing Technologies . Springer Series in Materials Science, vol 360. Springer, Cham. [Link]

M. A. Safaei et al., "Multiaxial Finite Strain Behavior of Polydomain Liquid Crystal Elastomers," *International Journal of Engineering Sciences*, under review. [Preprint Link]

M. A. Safaei et al., "Mechanical Characterization of Triply periodic Minimal Surfaces using Δ-PINNs," *in preparation*.

PROFESSIONAL EXPERIENCES

School of Mechanical Engineering, University of Tehran — Tehran, Iran

Research Assistant | **2022–Present**

- Supervise and mentor undergraduate students on research projects in computational mechanics, soft materials, and additive manufacturing.
- Coordinate and oversee part-time **3D-printing operations**, including student training and project support.
- Contributed as an author to Sections 1.3–1.5 of *Computational Modeling of Intelligent Soft Matter* [Link].

Teaching Assistant: Finite Element Methods [Link], Continuum Mechanics, Nonlinear Solid Mechanics [Link], and Research Methodology [Link].

Assistant Peer Reviewer (Volunteer) | **2023–Present**

Assisted with the peer-review process for **50+ manuscripts** submitted to journals including *Materials & Design*, *Thin-Walled Structures*, *Chemical Engineering Journal*, *Additive Manufacturing*, and *International Journal of Biological Macromolecules*.

Faradars Online Learning Platform — Remote, Iran

Course Instructor | **2024-Present**

- Developing online courses on **Smart Polymers [Link]** and **Neural Networks**.

Emamat High School — Tehran, Iran

English Language Instructor | **2024**

- Delivered general English instruction and prepared learning materials for students.

Nogolearn — Tehran, Iran

English Language Instructor (Upper-Intermediate/Advanced) | **2025**

- Taught upper-intermediate and advanced English courses with emphasis on academic communication and proficiency development.

IKCO Company — Tehran, Iran

Engineering Intern | **2021**

MDFICO — Tehran, Iran

Junior Engineering Intern | **2018–2020**

HONORS & AWARDS

Ranked 12th in the Nationwide PhD Entrance Examination (Solid Mechanics specialization).
Top 1%, out of more than 10,000 applicants in the M.Sc. Nation-wide University Entrance Exam (Full scholarship).
Top 5%, out of more than 190,000 applicants in the B.Sc. Nation-wide University Entrance Exam (Full scholarship).
Awarded within the top state candidates in the National Kharazmi Elite Youth Festival. (Poetry-Literature)

LICENCES & CERTIFICATES

Python: Fundamental & Advanced [Link] – COMSOL: Multiphysics [Link] – MATLAB: advanced – Excel – SEM & TEM [Link]

ACADEMIC PROJECTS

Artificial Intelligence / Neural Networks [Link]

- Implemented classical machine learning pipelines (KNN, SVM, Logistic Regression, k-means clustering).
- Constructed deep learning models (CNNs, MLPs, U-Net) for end-to-end image classification and semantic segmentation.
- Formulated reinforcement learning agents leveraging Monte Carlo methods and Deep Q-Networks (DQN).
- Engineered Tandem Neural Network (TNN) architectures for inverse design problems, predicting parameters of hyperelastic materials from target mechanical stress–strain responses [Link].
- Applied dimensionality reduction and clustering frameworks (PCA, UMAP) to classify features of structural lattices [Link].

Viscoelasticity [Link]

- Implemented Quasi-Linear Viscoelastic models to simulate benchmark problems including stress relaxation & creep response.
- Investigated the influence of glass transition temperature on rubbery plateau behavior using the WLF equation.
- Modeled the visco-hyperelastic behavior of hybrid TPMS lattice structures under thermomechanical loading conditions.

Nonlinear Solid Mechanics Course [Link]

- Developed apps & codes for material models calibration using multiaxial experimental data [Link].
- Formulated semi-analytical models for transient and quasi-static swelling of pH- and temperature-responsive hydrogels.
- Developed numerical and analytical models for thermal shape memory polymer (SMP) beams and lattice architectures.

FEM: Intro & Advanced Courses [Link]

- Developed UEL, UMAT, and UHYPER subroutines in ABAQUS for various hyperelastic constitutive models from the literature.
- Implemented the finite element method from scratch in MATLAB and Python for linear elasticity problems.

Bioliographic review [Link]

- Conducted a systematic bibliographic review and literature analysis on smart polymer composite actuators (or any given keyword).

References

- Mostafa Baghani - Professor – University of Tehran (Iran) - email
- Majid Baniassadi - Professor – University of Tehran (Iran) - email
- Mahdi Bodaghi – Associate Professor - Nottingham Trent University (UK) - email